

Singapore

Competition and Consumer Commission

Building upon its establishment of the Data and Digital ('D2') division in 2023, the Competition and Consumer Commission of Singapore ('CCCS') has continued to strengthen its technological capabilities in 2024 and 2025. This submission highlights three key initiatives that demonstrate CCCS's commitment to leveraging technology for more effective competition and consumer protection enforcement in digital markets.

I. AI Markets (AIM) Toolkit

CCCS launched the beta version of its AI Markets ('AIM') Toolkit in May 2025, marking a significant advancement in AI governance within Singapore's competition and consumer protection framework. Developed in partnership with the Infocomm Media Development Authority of Singapore ('IMDA'), AIM Toolkit integrates, as a specialized vertical, with the AI Verify Toolkit—an established open-source AI governance testing framework that aligns with international standards from the European Union, OECD, and Singapore.

AIM Toolkit employs a two-pronged approach to address AI deployment concerns in the marketplace. The first component comprises process check questions to help companies systematically evaluate both their AI models and deployment practices, enabling them to identify and address potential anti-competitive behaviors or unfair practices pre-emptively.

The second component builds on AI Verify's technical foundation and focuses on two critical aspects of AI governance—explainability and fairness. The explainability testing analyses AI model predictions through feature importance examination, while fairness testing evaluates prediction disparities across different groups using specific metrics. These technical assessments help ensure AI deployments maintain competitive neutrality and fairness to consumers in the marketplace.

The AIM Toolkit is designed to form part of CCCS's voluntary compliance programme for businesses. Adopting the toolkit AIM may be considered a mitigating factor should a business run afoul of competition law.

This initiative, building on the internationally recognized AI Verify framework, demonstrates CCCS's commitment to promoting responsible AI adoption while safeguarding market competition and consumer protection, and may serve as a reference for other competition authorities.

II. Merger Decision Bot

Merger Decision Bot is an AI assistant deployed in March 2025 and developed in-house by D2 on 'AIBots', an internal platform developed by Government Technology Agency ('GovTech') for Singapore government officers to create and use customised Generative AI chatbots.

This tool uses OpenAI's 'GPT-4o mini' AI model and leverages Retrieval-Augmented Generation ('RAG') and long context processing capabilities to provide context-aware responses with proper citations. Implemented with safety guardrails against harmful discourse and prompt-engineered to respond specifically to merger related queries, it accesses an extensive repository of CCCS's merger guidelines and merger decisions.

Since its deployment, Merger Decision Bot has enhanced CCCS's merger review capabilities, particularly benefiting new officers by enabling efficient access to institutional knowledge. It facilitates the retrieval of relevant precedents and assessment principles from past cases, supporting analytical consistency in merger control. This initiative demonstrates CCCS's commitment to incorporating AI technology while maintaining rigorous regulatory standards. With the experience gained from this successful implementation, CCCS is well-positioned to extend AI assistants to other areas of knowledge management in the future.

III. A/B Testing Platform

CCCS has developed an in-house A/B testing platform to empirically assess the impact of online choice architecture on consumer behavior. This development comes at a critical time when business transactions are increasingly conducted through digital platforms, making it essential to understand how interface design and choice presentation influence consumer behavior and decisions.

Launched in December 2024, this platform provides a user-friendly environment for conducting controlled experiments. Its low-code interface allows investigators to create test websites that replicate specific choice architectures under investigation. The platform automatically generates comprehensive result

summaries, measuring key metrics such as click-through and conversion rates to quantify the effects of various digital interface designs.

The development process involved extensive consultation with both domestic government agencies and international competition authorities with experience in empirical studies, ensuring that the platform incorporated best practices and lessons learned from similar initiatives worldwide.

Looking ahead, CCCS is pursuing a research collaboration with Singapore's Civil Service College to investigate the influence of dark patterns on Singaporean consumers' behavior. This partnership aims to translate research findings into practical policy-making reference materials, strengthening CCCS's evidence-based approach to regulation and policy development in digital markets.